

Existing Problem

Heuristics-Based Methods:

Qiskit/Tket... Fixed sequence of passes and heuristic

Not effective for arbitrary circuits; no guarantee for coverage

Search-Based Methods:

Queso/Guoco/Quartz Auto-synthesize thousands of rewrite rules
Apply exploration/superoptimization algorithm

Large set of rules; limited expressiveness; limited symbolic rewrite

Our Goal

We want to synthesize

- 1) **symbolic rewrite** that able to capture all possible common patterns of a rewrite
- 2) **small** set of rules that proven complete within a bound
- 3) rules are **effective** in optimizations